10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 12 — Waves

Solved Exercise | New Book (PECTAA 2026)

Exercise 12 (Solved)

This exercise covers the nature of waves, the transfer of energy, mechanical and electromagnetic waves, transverse and longitudinal waves, and the behaviour of waves at a boundary — reflection, refraction and diffraction.

A. Multiple Choice Questions

12.1 The direction of vibration in a longitudinal wave is:

(a) at right angles to the wave direction
(b) in circular motion
(c) opposite to the wave direction
(d) in the same direction as the wave travels

Answer: (d) in the same direction as the wave travels — in a longitudinal wave the particles vibrate to and fro along the same line in which the wave moves, forming compressions and rarefactions.

12.2 Which part of a transverse wave carry the lowest point?

(a) Crest
(b) Compression
(c) Rarefaction
(d) Trough

Answer: (d) Trough — in a transverse wave the highest point is the crest and the lowest point is the trough; compression and rarefaction belong to longitudinal waves.

12.3 Which of the following wave type requires a medium to travel?

(a) X-rays
(b) Light waves
(c) Radio waves
(d) Sound waves

Answer: (d) Sound waves — sound is a mechanical wave, so it needs a material medium (air, water, solid); X-rays, light and radio waves are electromagnetic and need no medium.

12.4 The property of a wave which indicates how much energy it carries is:

(a) crest
(b) amplitude
(c) wavelength
(d) speed

Answer: (b) amplitude — the energy of a wave is directly proportional to the square of its amplitude, so a wave of larger amplitude carries more energy.

12.5 A wave front is:

(a) the distance between two crests
(b) a region of high pressure in a wave
(c) a line joining points that vibrate in phase
(d) the outer edge of a wave medium

Answer: (c) a line joining points that vibrate in phase — all the points on one wave front are at the same stage of vibration, i.e. in the same phase.

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12.6 The wave phenomenon which occurs when waves change direction due to a change in speed is:

(a) diffraction
(b) reflection
(c) absorption
(d) refraction

Answer: (d) refraction — when a wave passes from one medium to another its speed changes, and this change of speed bends its path — that is refraction.

12.7 What happens to wave speed and wavelength when a tsunami approaches shallow water?

(a) Speed increases, wavelength increases
(b) Speed decreases, wavelength shortens
(c) Speed remains the same, wavelength shortens
(d) Speed increases, wavelength stays constant

Answer: (b) Speed decreases, wavelength shortens — in shallow water the wave slows down; the frequency stays the same, so $\lambda = v / f$ becomes shorter and the height of the wave rises.

12.8 Which wave behaviour explains how sound is heard around a corner?

(a) Reflection
(b) Refraction
(c) Interference
(d) Diffraction

Answer: (d) Diffraction — when the wavelength is comparable with the size of an obstacle, the wave bends around it; sound waves bend round the edge of a wall, so we hear the person speaking behind it.

B. Short Answer Questions

12.1 What do waves transfer from one place to another?

Waves transfer energy and momentum from one place to another, not matter. The particles of the medium only vibrate about their mean positions and do not move along with the wave.

For example, when a stone is dropped in water, the ripples carry energy outwards, but the water itself does not travel outward. In the same way sound carries energy through air to our ears, and light carries energy from the Sun to the Earth.

12.2 Name any two examples of mechanical waves.

Two examples are sound waves, which travel through air, water or solids, and water waves (ripples) on the surface of a liquid.

Waves on a stretched string, seismic waves produced by an earthquake and the waves of a spring (slinky) are also mechanical waves, because all of them need a material medium.

12.3 Which type of wave can travel through vacuum?

Electromagnetic waves can travel through a vacuum. They do not need any material medium because they are produced by oscillating electric and magnetic fields.

Light, radio waves, microwaves, infrared, ultraviolet, X-rays and gamma rays are all electromagnetic waves. This is why sunlight and radio signals reach us through

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space, while sound cannot.

12.4 What is the difference between transverse and longitudinal waves?

Point	Transverse wave	Longitudinal wave
Direction of vibration	at right angles to the direction of the wave	along the same line as the direction of the wave
Shape of the wave	crests and troughs	compressions and rarefactions
Regions formed	high parts (crests) and low parts (troughs)	crowded parts (compressions) and spread-out parts (rarefactions)
Can it be polarised?	yes	no
Examples	light, radio waves, waves on a string, water ripples	sound waves, waves in a spring pushed and pulled along its length

12.5 What is meant by wavelength?

Wavelength is the distance between two consecutive points of a wave which are in the same phase — for example, the distance from one crest to the next crest, or from one trough to the next trough.

In a longitudinal wave it is the distance between the centres of two consecutive compressions or two consecutive rarefactions. It is denoted by the Greek letter λ (lambda) and measured in metres. It is related to speed and frequency by $v = f\lambda$.

12.6 Write the definition of frequency.

Frequency is the number of complete vibrations (or cycles) of a wave passing a fixed point in one second. Its unit is the hertz (Hz), where 1 Hz = 1 vibration per second.

It is the reciprocal of the time period: $f = 1 / T$. A wave of frequency 50 Hz completes 50 vibrations in one second, and its time period is 0.02 s.

12.7 What causes a tsunami in the ocean?

A tsunami is caused by a sudden, large disturbance of the sea floor or of the ocean water. Usually this is a strong undersea earthquake where one plate of the Earth's crust suddenly slips under or past another and lifts a huge section of the ocean floor, pushing the water above it upward. Underwater volcanic eruptions, landslides and, rarely, meteorite impacts can also cause a tsunami.

In the deep ocean the wave has a very long wavelength (often hundreds of kilometres) and a small height, and it travels very fast, sometimes 700–800 km h^{-1}. As it reaches shallow coastal water it slows down, the wavelength shortens and the energy is squeezed into a smaller depth, so the wave rises into a wall of water several metres high and floods the coast with great force.

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C. Constructed Response Questions

12.1 Why sound cannot travel through space, but light can? Explain using the concept of wave types.

Sound is a mechanical wave. It is produced by the vibration of particles of a material medium, and it travels forward as a series of compressions and rarefactions passed from one particle to the next. If there are no particles, there is nothing to compress, so no wave can be formed or carried. Space is almost a complete vacuum, with hardly any particles, therefore sound cannot travel through it — that is why the Universe is silent and why astronauts cannot talk to each other directly in space.

Light is an electromagnetic wave. It is made of oscillating electric and magnetic fields which do not need any material particles; the fields themselves move forward. Because of this, light can travel through a vacuum at 3×10^8 m s^{-1}. That is how the light and heat of the Sun reach the Earth across 150 million kilometres of empty space, and how radio waves from satellites reach our dishes.

12.2 A student shakes one end of a rope and observes waves move along it. What kind of wave is this, and how do the particles move?

This is a transverse wave, because the rope is shaken up and down while the wave travels along the length of the rope — the vibration is at right angles to the direction of travel.

The particles of the rope do not move along the rope with the wave. Each particle simply vibrates up and down about its own fixed mean position and passes the disturbance on to the next particle. The crests and troughs move forward, but the material of the rope only oscillates in place, so energy travels along the rope while matter does not. If a ribbon is tied to the middle of the rope, it only moves up and down and stays at the same place.

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12.3 How does the amplitude of a wave relate to the energy it carries? Give an example to support your answer.

The energy carried by a wave is directly proportional to the square of its amplitude ($E \propto A^2$). If the amplitude is doubled, the energy carried becomes four times as large. Amplitude therefore tells us how much energy the wave is carrying.

Example: when a stone is dropped gently into a pond, the ripples are low (small amplitude) and die out quickly; a heavy stone thrown hard makes tall ripples which travel far and can rock a toy boat. In the same way a loud sound has a large amplitude and a soft sound a small one: the loudspeaker cone vibrates with a bigger amplitude and sends out much more energy, which is why a very loud sound can damage the ear.

12.4 Why do we still hear a person talking even if he is behind a wall? Explain using the concept of diffraction.

Diffraction is the spreading out of waves when they pass through a gap or around the edge of an obstacle. A wave diffracts noticeably only when the size of the gap or obstacle is comparable with the wavelength of the wave.

The sound of a person talking has a wavelength of the order of 1 m (about 0.3 m to 3 m), which is close to the size of a door, a window or the corner of a wall. So the sound waves readily bend around the corner or spread through the doorway and reach our ears, and we hear the person even though we cannot see him. Light, on the other hand, has a wavelength of about 5×10^{-7} m, which is far smaller than the wall, so it hardly bends round it and we cannot see the speaker.

12.5 If size of slit, a wave passes through is much smaller than the wavelength, what will happen to the wave after passing through the slit? Explain briefly.

When the slit is much smaller than the wavelength, the wave behaves almost like a point source at the slit and spreads out widely on the other side in nearly circular (spherical) wave fronts. The diffraction is very strong and the wave energy is scattered in almost all directions, so no sharp beam or shadow is formed.

The amplitude of the diffracted wave is very small, so the intensity of the sound or light beyond the slit becomes weak. For example, a narrow doorway diffracts sound strongly in all directions, so the sound is heard everywhere but it is faint; and light passing through a very narrow hole spreads out and forms a broad, dim patch instead of a sharp image.

D. Comprehensive Questions

12.1 Explain how waves transfer energy without transferring matter. Give examples from daily life.

A wave is a disturbance that travels through a medium because the particles of the medium pass on the disturbance to their neighbours. Each particle vibrates about its mean position, pushing or pulling the next particle and then returning, so at the end of the vibration the particle is back where it started. No particle is carried along with the wave from one place to another; only the disturbance, and with it the energy, moves forward.

The energy is carried because work is done on each successive particle. In a sound wave the particles of air oscillate and pass the energy to the next layer of air until it reaches the ear drum; in a water wave the water particles move up and down in circles while the pattern of the wave travels across the surface; in a wave on a rope the rope particles only oscillate up and down.

Everyday examples: a cork floating on water bobs up and down as ripples pass but does not travel with them; a crowd doing the 'wave' in a stadium stays seated in the same seats while the pattern moves round the ground; a football crowd's shout travels across the field as a sound wave without any air being carried along; and the Sun's energy reaches us as electromagnetic waves with no matter travelling from the Sun.

12.2 Describe the difference between mechanical and electromagnetic waves with suitable examples.

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Point	Mechanical waves	Electromagnetic waves
Definition	waves that need a material medium for their propagation	waves produced by oscillating electric and magnetic fields
Medium	necessary — solid, liquid or gas	not needed; can travel through a vacuum
Speed	different in each medium; sound in air ≈ 340 m s^{-1}	all travel at 3 × 10^{8} m s^{-1} in vacuum (light)
Nature	may be transverse (rope, light-like water waves) or longitudinal (sound)	always transverse
Examples	sound waves, water waves, waves on a string, seismic waves, waves in a spring	light, radio waves, microwaves, infrared, ultraviolet, X-rays, gamma rays

In short, mechanical waves are the vibration of matter passed on from particle to particle, while electromagnetic waves are vibrations of fields which can cross empty space. This is why sound cannot be heard in space but sunlight can reach us.

12.3 What are transverse and longitudinal waves? Explain each with the help of diagrams and examples.

Transverse wave. In a transverse wave the particles of the medium vibrate at right angles to the direction in which the wave travels. The wave has high points called crests and low points called troughs, and the wavelength is the distance between two consecutive crests.

Picture a rope tied at one end: if the free end is shaken up and down, a hump travels along the rope while the rope itself only moves up and down. Light, radio waves, water ripples and waves on a string are transverse waves.

Longitudinal wave. In a longitudinal wave the particles vibrate to and fro along the same line in which the wave travels. The wave consists of compressions, where the particles are crowded together, and rarefactions, where they are spread apart; the wavelength is the distance between two consecutive compressions.

Picture a slinky spring: when its end is pushed and pulled quickly along the length of the spring, crowded and spread-out coils travel along it. Sound waves in air, waves in a spring and seismic P-waves are longitudinal waves. (Diagrams: draw crests and troughs for the transverse case and evenly spaced plus crowded dots for the longitudinal case.)

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12.4 Discuss the properties of waves; such as reflection, refraction, and diffraction. Support your answer with examples.

Reflection. When a wave strikes a boundary between two media, part or all of it bounces back into the same medium. The angle of incidence is equal to the angle of reflection, and the wavelength, frequency and speed do not change. Example: echo of sound from a hill; light reflected from a mirror; water waves bouncing back from the wall of a tank.

Refraction. When a wave passes from one medium into another, its speed changes and, if it enters at an angle, its direction changes too. Its frequency stays the same but its wavelength changes according to $\lambda = v / f$. Example: a pencil looks bent at the surface of water; sound travels faster in warm air than in cold air and bends upward on a hot day; ocean waves slow down and bend as they come into shallow water, so they seem to turn towards the shore.

Diffraction. This is the spreading of waves after passing through a gap or around an obstacle. It is strong when the gap is of the same size as, or smaller than, the wavelength. Example: sound from a room reaches us through a doorway and around corners; radio waves of long wavelength bend round hills and buildings; light through a narrow slit produces a pattern of fringes.

So reflection keeps a wave in the same medium, refraction changes its speed and direction, and diffraction makes it spread out and bend round obstacles.

12.5 What is diffraction? How do wavelength and slit size affect the diffraction of waves?

Diffraction is the bending or spreading of waves when they pass through an opening or around the edge of an obstacle. Because of diffraction, waves do not stop sharply at the edge of a gap but spread into the region behind it.

Effect of slit size. When the width of the slit is much larger than the wavelength, the wave passes almost straight through and only the edges are slightly bent, so diffraction is small. When the slit is of the same order as the wavelength, the wave spreads out clearly on the other side in a fan shape. When the slit is much smaller than the wavelength, the wave spreads out almost equally in all directions and the intensity becomes weak.

Effect of wavelength. For a given slit, a longer wavelength diffracts more than a shorter one. That is why long-wavelength sound bends round a doorway and can be heard round the corner, while light of wavelength 5×10^{-7} m does not bend noticeably round the same doorway. Long radio waves are also able to bend round the hills and reach far villages.

E. Numerical Problems

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12.1 The wavelength of a wave is 2 metres and its frequency is 5 Hz. What is the speed of the wave?

Given: $\lambda = 2$ m
$f = 5$ Hz

To find: $v = ?$

Formula: $v = f\lambda$

Solution: $v = 5 \times 2$
$v = 10 \text{ m s}^{-1}$

Answer: $v = 10 \text{ m s}^{-1}$

12.2 A wave travels at a speed of 300 m s^{-1} and has a wavelength of 3 metres. Calculate its frequency.

Given: $v = 300 \text{ m s}^{-1}$
$\lambda = 3$ m

To find: $f = ?$

Formula: $f = v / \lambda$

Solution: $f = 300 / 3$
$f = 100$ Hz

Answer: $f = 100$ Hz

12.3 A sound wave has a frequency of 250 Hz and a speed of 340 m s^{-1}. Find its wavelength.

Given: $f = 250$ Hz
$v = 340 \text{ m s}^{-1}$

To find: $\lambda = ?$

Formula: $\lambda = v / f$

Solution: $\lambda = 340 / 250$
$\lambda = 1.36$ m

Answer: $\lambda = 1.36$ m

12.4 A wave has a frequency of 50 Hz. How much time does it take to complete one wave?

Given: $f = 50$ Hz

To find: $T = ?$

Formula: $T = 1 / f$

Solution: $T = 1 / 50$
$T = 0.02$ s

Answer: $T = 0.02$ s

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12.5 A wave has a wavelength of 0.5 m and frequency of 200 Hz. Calculate the speed of the wave.

Given: $\lambda = 0.5$ m
$f = 200$ Hz

To find: $v = ?$

Formula: $v = f\lambda$

Solution: $v = 200 \times 0.5$
$v = 100 \text{ m s}^{-1}$

Answer: $v = 100 \text{ m s}^{-1}$

12.6 The time period of a wave is 0.02 seconds. Calculate the frequency of the wave.

Given: $T = 0.02$ s

To find: $f = ?$

Formula: $f = 1 / T$

Solution: $f = 1 / 0.02$
$f = 50$ Hz

Answer: $f = 50$ Hz

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Points to Remember

- Waves transfer energy, not matter.
- $v = f\lambda$, $f = 1 / T$, energy $\propto$ (amplitude)^2.
- Mechanical waves need a medium; electromagnetic waves do not.
- Transverse: vibration at right angles; longitudinal: vibration along the wave.
- Reflection (bounce), refraction (speed change), diffraction (spreading at a gap).